# Week 7

## Student Performance – Data Cleaning & Transformation

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### 1. Introduction

The Week 7 project focuses on Student Performance – Data Cleaning and Transformation using Microsoft Power BI and Power Query. The main purpose of this project is to clean, organize, and transform the student performance dataset so that it can be used for further analysis.

In this project, I imported the student performance dataset, inspected and renamed the columns, created Total Score and Average Score, classified students using Grade and Pass/Fail, performed Unpivot on the subject scores, and rounded the Average Score. This project helped me understand how raw data can be transformed into a structured and analysis-ready dataset using Power Query.

### 2. Import the Dataset

I imported the Student Performance in Exams dataset into Power BI using the Text/CSV option. After importing the dataset, I selected Transform Data to open the Power Query Editor.

StudentsPerformance.csv

File Origin: 1252: Western European (Windows)
Delimiter: Comma
Data Type Detection: Based on first 200 rows

| gender | race/ethnicity | parental level of education | lunch | test preparation course | math score | reading score | writing score |
|---|---|---|---|---|---|---|---|
| female | group B | bachelor's degree | standard | none | 72 | 72 | 74 |
| female | group C | some college | standard | completed | 69 | 90 | 88 |
| female | group B | master's degree | standard | none | 90 | 95 | 93 |
| male | group A | associate's degree | free/reduced | none | 47 | 57 | 44 |
| male | group C | some college | standard | none | 76 | 78 | 75 |
| female | group B | associate's degree | standard | none | 71 | 83 | 78 |
| female | group B | some college | standard | completed | 88 | 95 | 92 |
| male | group B | some college | free/reduced | none | 40 | 43 | 39 |
| male | group D | high school | free/reduced | completed | 64 | 64 | 67 |
| female | group B | high school | free/reduced | none | 38 | 60 | 50 |
| male | group C | associate's degree | standard | none | 58 | 54 | 52 |
| male | group D | associate's degree | standard | none | 40 | 52 | 43 |
| female | group B | high school | standard | none | 65 | 81 | 73 |
| male | group A | some college | standard | completed | 78 | 72 | 70 |
| female | group A | master's degree | standard | none | 50 | 53 | 58 |
| female | group C | some high school | standard | none | 69 | 75 | 78 |
| male | group C | high school | standard | none | 88 | 89 | 86 |
| female | group B | some high school | free/reduced | none | 18 | 32 | 28 |
| male | group C | master's degree | free/reduced | completed | 46 | 42 | 46 |
| female | group C | associate's degree | free/reduced | none | 54 | 58 | 61 |

The data in the preview has been truncated due to size limits.

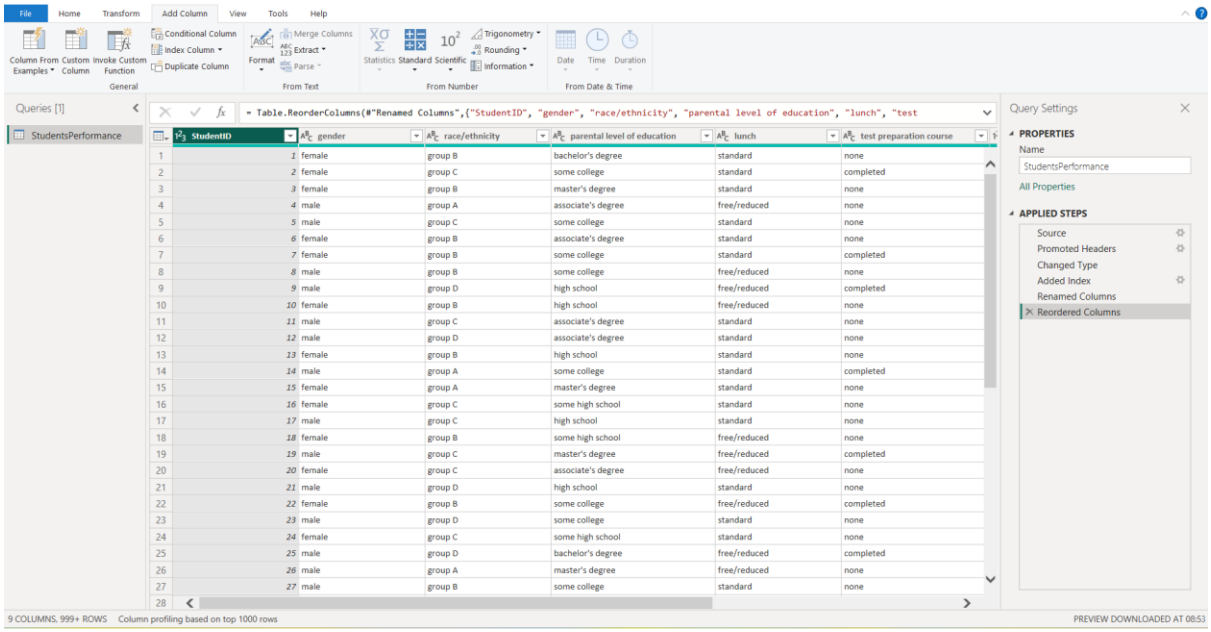
Extract Table Using Examples | Load | Transform Data | Cancel

### 3. Explore and Inspect the Data

I checked the columns and records available in the dataset. The dataset contains student information such as:

- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course
- Math Score
- Reading Score
- Writing Score

I also added an Index Column and named it StudentID.

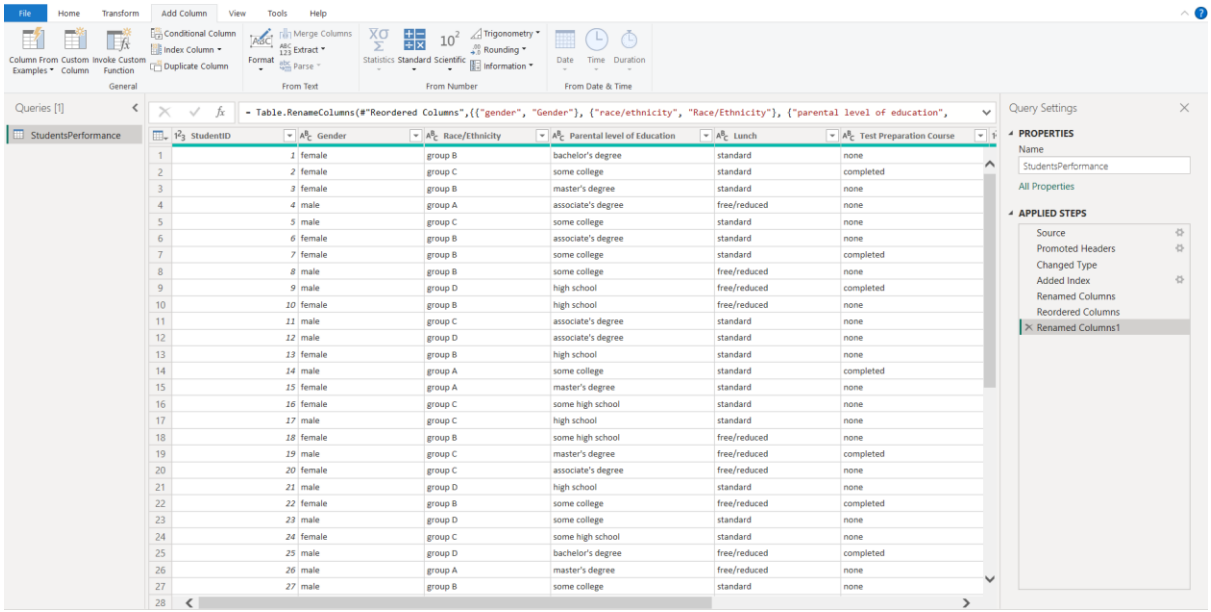



## 4. Rename the Columns

I renamed the columns to make them easier to understand and use.

For example:

- gender → Gender
- race/ethnicity → Race/Ethnicity
- parental level of education → Parental Level of Education
- lunch → Lunch
- test preparation course → Test Preparation Course
- math score → Math Score
- reading score → Reading Score
- writing score → Writing Score



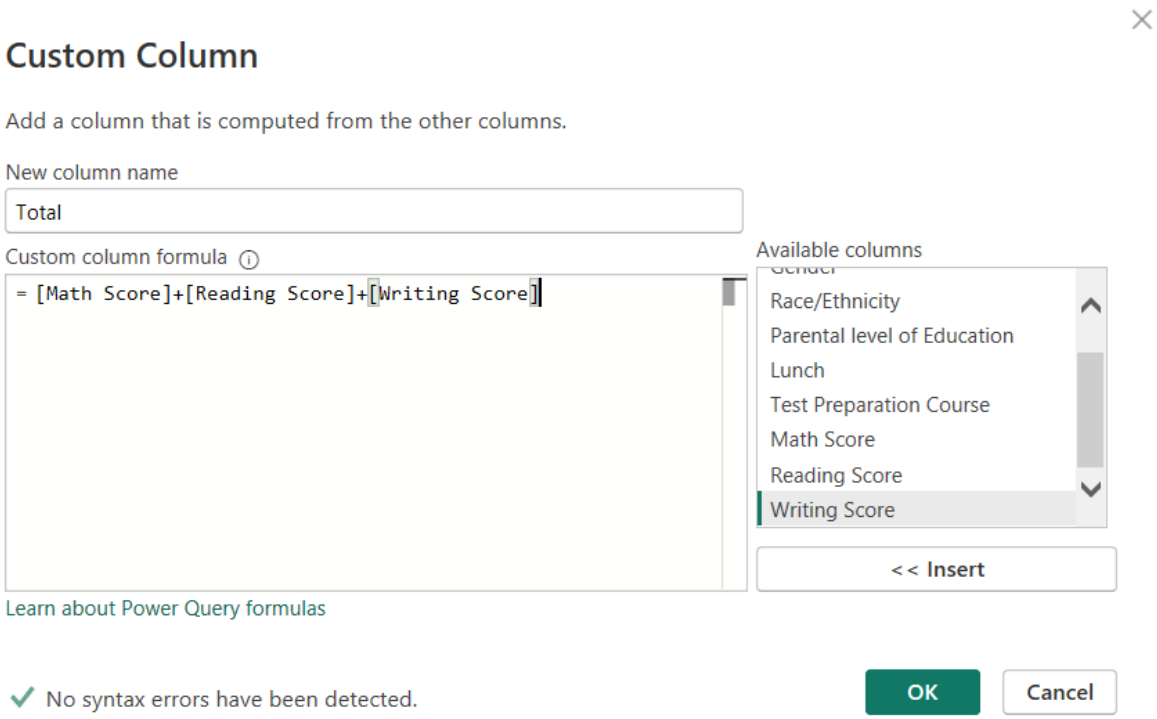

## 5. Create Total Score

I created a Custom Column named Total.

The formula used was:

Total = Math Score + Reading Score + Writing Score

This calculates the total marks obtained by each student in all three subjects.



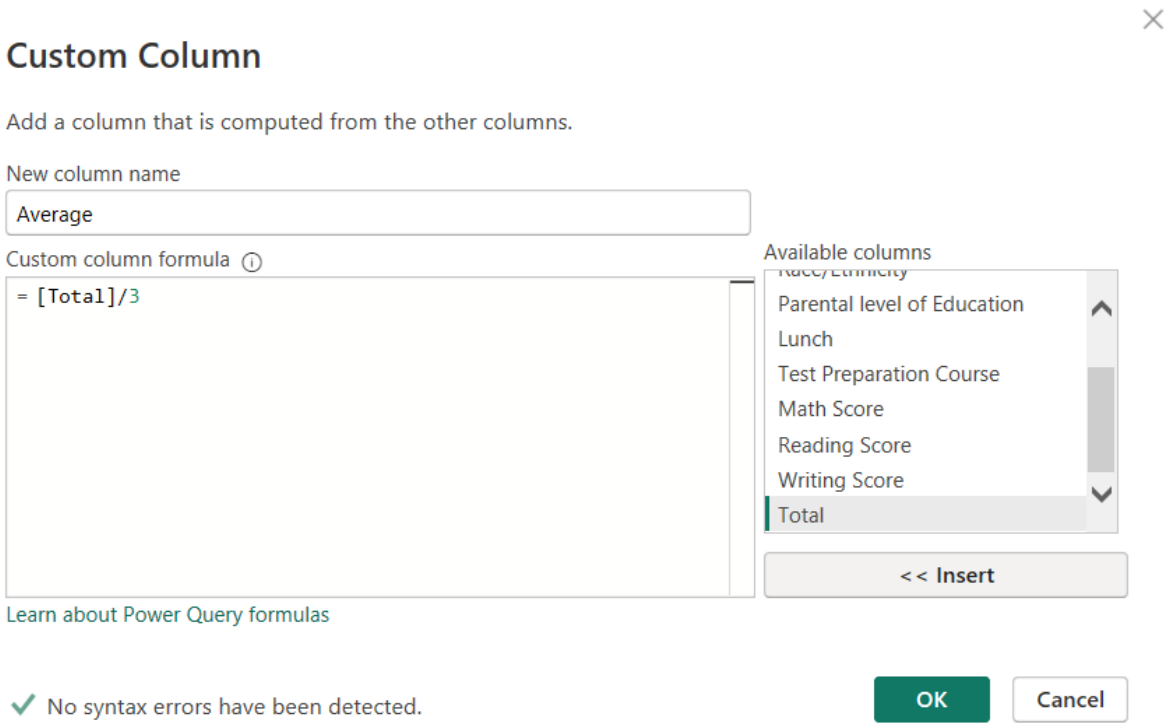

**6. Create Average Score**

I created another Custom Column named Average.

The formula used was:

Average = Total / 3

This calculates the average marks of each student across Math, Reading, and Writing.



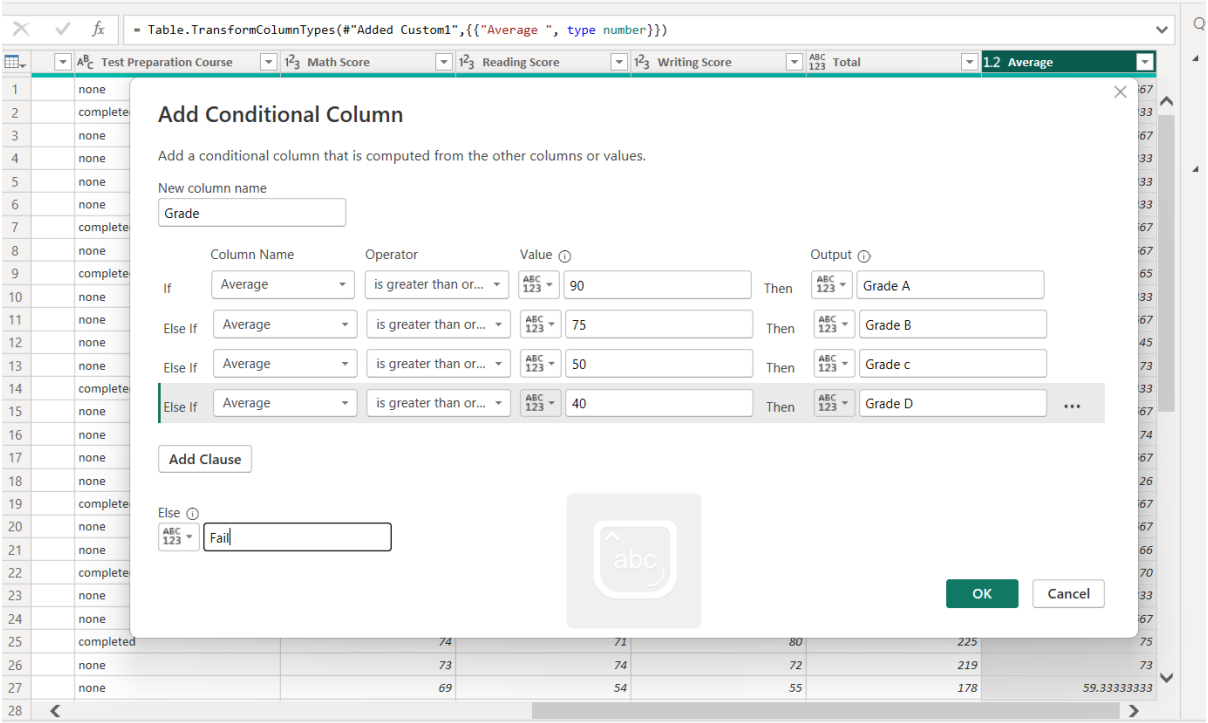

### 7. Create Grade

I created a Conditional Column named Grade based on the Average score.

The conditions used were:

- Average ≥ 90 → Grade A
- Average ≥ 75 → Grade B
- Average ≥ 50 → Grade C
- Average ≥ 40 → Grade D
- Below 40 → Fail

This helps to classify students according to their overall performance.




## 8. Create Pass/Fail

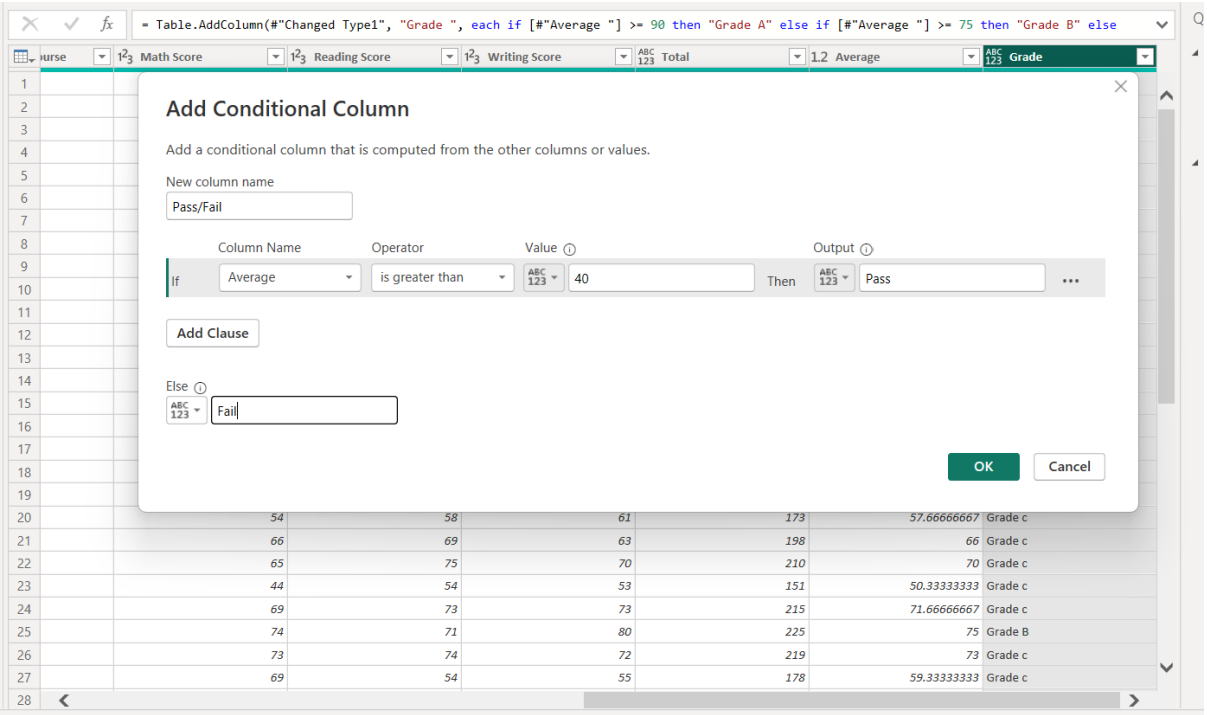
I created another Conditional Column named Pass/Fail.

The condition used was:

Average > 40 → Pass

Otherwise → Fail

This classifies students based on the defined passing criteria.



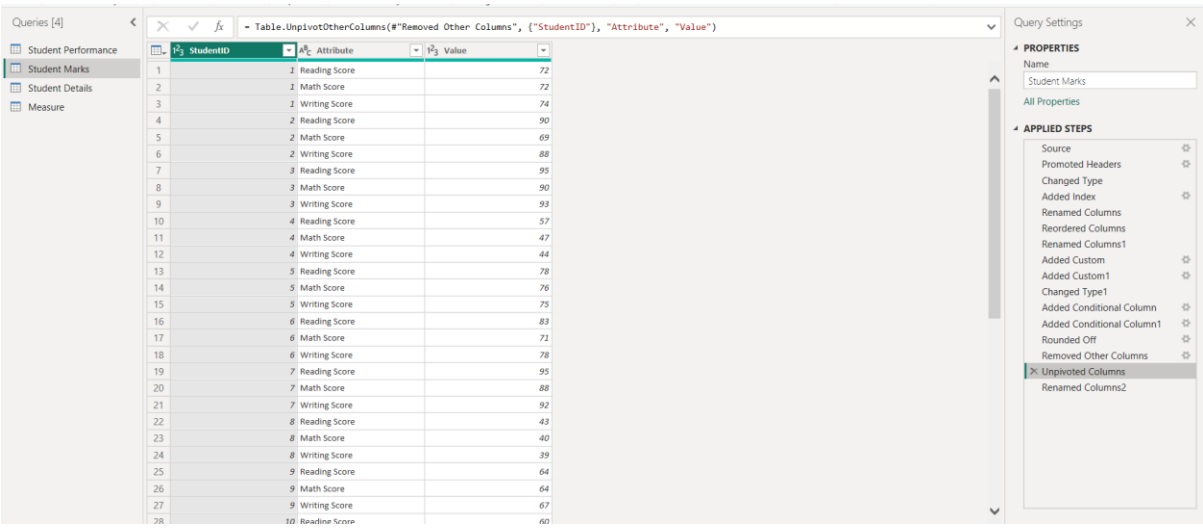

## 9. Unpivot Subject Scores

I selected the Math Score, Reading Score, and Writing Score columns and used the Unpivot Columns option in Power Query.

After unpivoting, the subject scores were transformed into two columns:

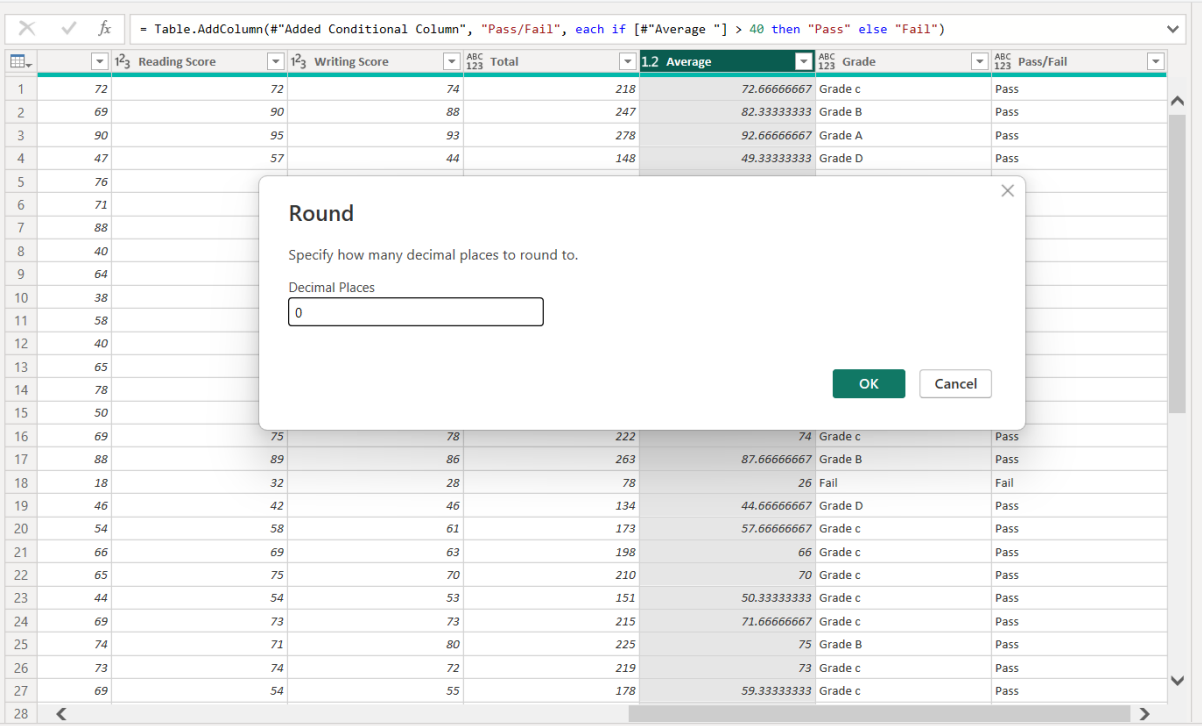
- Subject – contains Math, Reading, and Writing
- Score – contains the corresponding marks

This transformation makes it easier to analyze and compare student performance across different subjects.




## 10. Round the Average Score

The Average column contained decimal values. I used the Round option and selected 0 decimal places to make the values easier to read.



## 11.Creating DAX Measures

- Average Math Score

  Average Math Score = AVERAGE('Student Performance'[Math Score])

- Average Reading Score

  Average Reading Score = AVERAGE('Student Performance'[Reading score])

- Average Writing Score

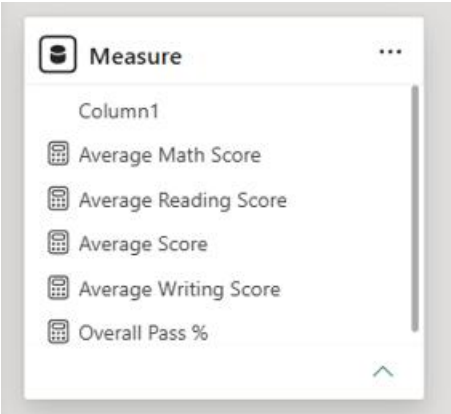
  Average Writing Score = AVERAGE('Student Performance'[Writing score])

- Overall Average Score

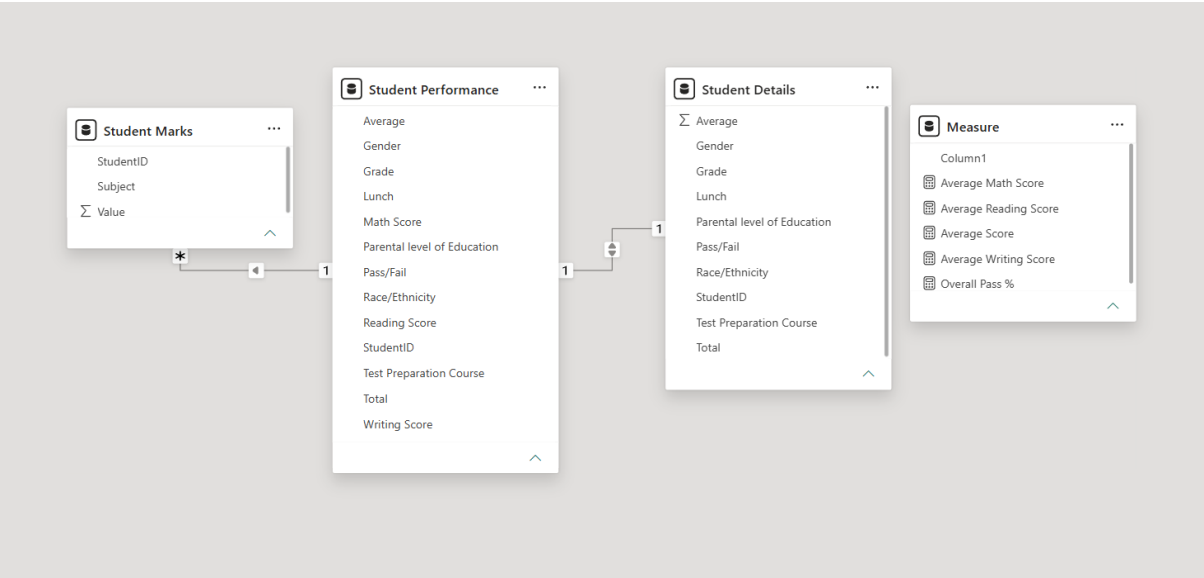
  Overall Pass % =

  ```
  DIVIDE(
    CALCULATE(
      COUNTROWS('Student Performance'),
      'Student Performance'[Pass/Fail] = "Pass"
    ),
    COUNTROWS('Student Performance'),
    0
  )
  ```

- Overall Pass %




## 12.Final Data Model Relationship




## 13.Conclusion

In this Week 7 project, I used Power BI Power Query to transform the Student Performance dataset. I added StudentID, reordered and renamed columns, created Total and Average scores, created Grade and Pass/Fail classifications, unpivoted the subject scores, and rounded the Average Score.

This project helped me understand how to use Power Query for data cleaning and transformation and how to prepare data for further analysis and visualization in Power BI.